



ICT 1201

Fundamentals of Computer Systems

Introduction

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Learning Outcomes

- Describe basic concepts related to computer and information technology
- Use popular application software used for word processing, spreadsheet calculation, office database, presentations and graphics processing.
- Describe the functions of popular operating systems and use a command line interface to interact with an operating system.

Learning Outcomes.....

- Describe the functions of the peripherals of a computer.
- Demonstrate how to assemble a computer.

Syllabus

Introduction	• Early history of computing • The computer generations • Characterization • Classification of computers • Basic components and organization of a computer • Representation of information in a computer • Concept of programming and programming languages • Language translation
Application Software	• Basic features of application software • Tools for work and study • Desktop Accessories • Word Processing • Spreadsheets • Database Management Systems • Graphics • Communication • Software suites

Syllabus.....

Operating Systems and its functions	• The need for an operating system • The types of operating systems • The features of the MS-DOS / Windows XP operating system, Linux • Utility programs • Utility packages
Peripherals	• Input devices and its functions • Output devices and its functions • Processing and memory hardware • Secondary storage and communication devices • Introduction to device ports • Assembling a computer
Computers and Communication	• Revolution in computers and communication

Syllabus.....

The digital future	• Role of Information Technology in society • Distinguish between data and information • Properties of information and basic IT tools • e-learning • e-banking
Social issues	• Ethics and standards in computing • Copyright • Intellectual Property Right • Piracy

Evaluation Criteria

- Continuous Assessment:
 - Theory tests (in class, Assignments)
 - Practical tests
 - Mini Project

- End Semester Examination:
 - Practical
 - Theory

References

- Introduction to Computer Science by Ramon A. Mata Toledo, Paulink K. Cushman
- Introduction to Computers by Peter Norton

Early History of Computing

What is a Computer ?

- A computer is an electronic device that process data, converting it into information that is useful to people.
- Any computer regardless of its type is controlled by programmed instructions, which give the machine a purpose and tell it what to do.

- Imagine that you have a job where you have to perform nothing but calculations every day.
- At one point of time you will get bored, become careless, slow and start making mistakes.
- Therefore, inventors have been searching for a way to automate the task of performing calculations.

Pre Mechanical Computing

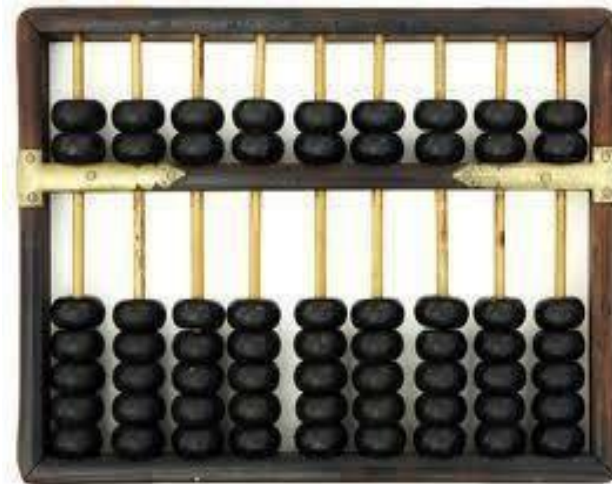


Early computational devices

Mechanical Computing

□ Abacus

- Chinese invented the Abacus in 3000 BC as the first mechanical computing tool.
- Addition, subtraction, multiplication and division are performed by moving the appropriate beads according to rules.



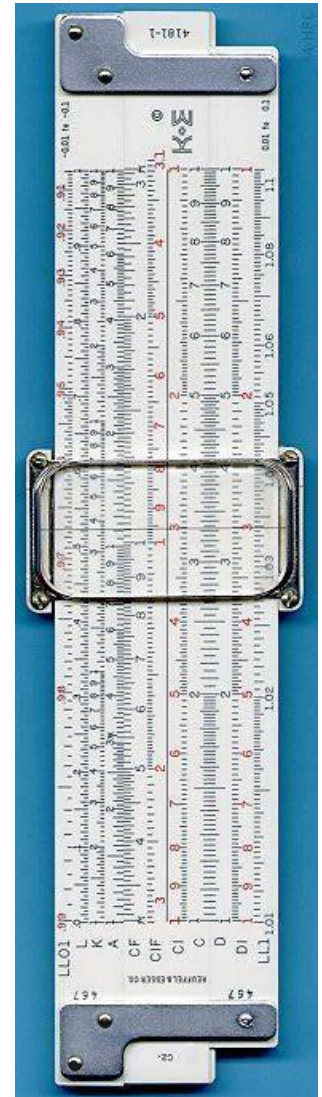
□ Napier's Bones

- In 1617 John Napier invented Napier's bones and logarithms.
- Using the multiplication tables embedded in the rods, multiplication can be reduced to addition operations and division to subtractions.



❑ Slide Rule

- William Oughtred made the first slide rule.
- Used for multiplication, division, trigonometry, squares and square roots, cubes and cube roots and logarithms as well.
- Before the invention of the pocket or handheld calculator the slide rule was a popular tool for calculations. (until 1974)



❑ Pascaline

- In 1642, French mathematician **Blaise Pascal**, invented the **Pascaline** – **first mechanical adding machine.**
- Wooden box that could only add and subtract numbers by means of a series of gears and wheels.



Inside of Pascaline

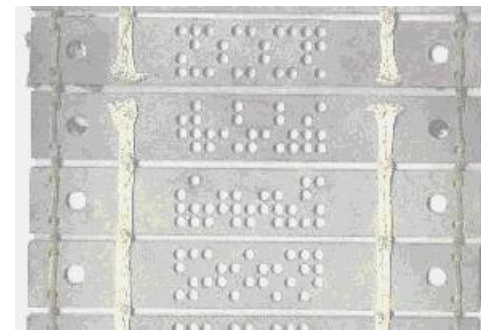
□ Leibniz

- In 1673, German inventor **Gottfried Leibniz** invented Leibniz Calculator known as the **Stepped Reckoner**.
- A calculating machine, but much superior to that of the Pascaline.
- Other than addition and subtraction, multiplication, division and find square roots of numbers can be done.



❑ Jacquard loom

- In 1801, **Jacquard** invented the **Jacquard loom**.
- It was a weaving machine that was controlled by **punch cards**.
- Made cloths faster and better than they could by hand.
- Punch cards are used for music to be played by **automatic pianos** and the **storing of programs** for computers.



❑ Difference Engine

- In 1822, Charles Babbage, an English mathematician, designed a machine called the Difference Engine to calculate the mathematical tables, such as logarithm tables.

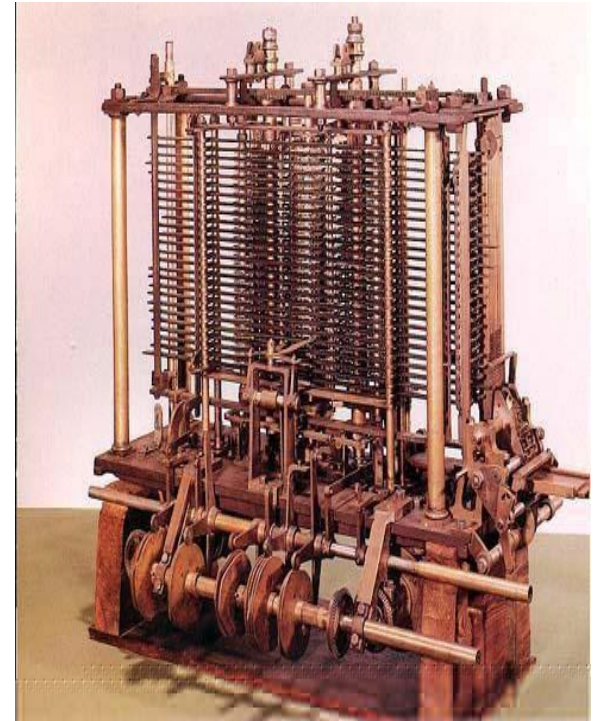


❑ Analytical Engine

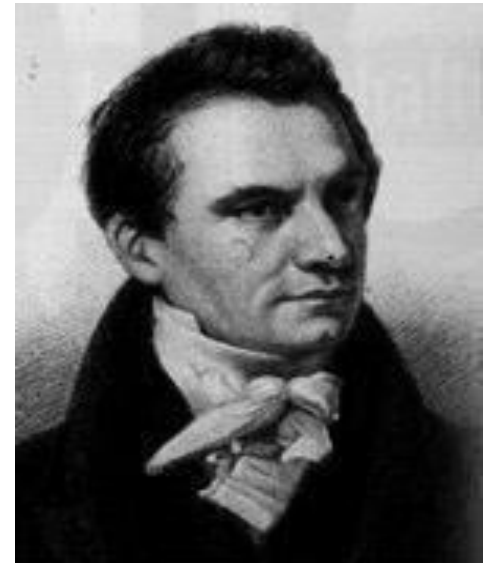
- In the 1830s, Babbage designed a second computing machine called the Analytical Engine.
- This machine was to be used in calculating complicated problems by following a set of instructions.
- He realized that punch cards could be employed as a storage mechanism, holding computed numbers for future reference.

- Since it uses punch-cards similar to those used by Jacquard loom he called the two main parts of his Analytical Engine the,
Store - where numbers were held and
Mill-where they were woven into new results, as
both terms are used in the waving industry.
- In a modern computer these same parts are called the
memory unit and central processing unit.

- The Analytical Engine also had a key function that distinguishes computers from calculators – the conditional statement.
- Based on the conditional statement the path of the program can be determined based upon a situation that is detected at the very moment the program is running.



- The two machines were unable to be finished because of the lack of funding and the failure of the advanced technology at that time.
- But computers after 1900 to today influence the technologies of those two.
- Therefore Charles Babbage is known as the **Father of the computer.**



- Augusta Ada Byron helped in programming in Charles Babbage's Analytical Engine.
- Suggested how to work analytical engine and the ideas of fundamentals of programming.
- She invented the subroutine and was the first to recognize the importance of looping.
- Ada known as the first computer programmer.

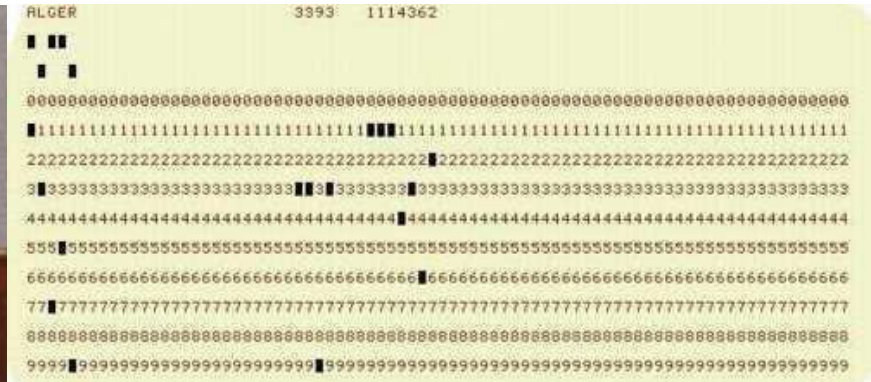


Electro - Mechanical Computing

❑ Hollerith tabulating machine

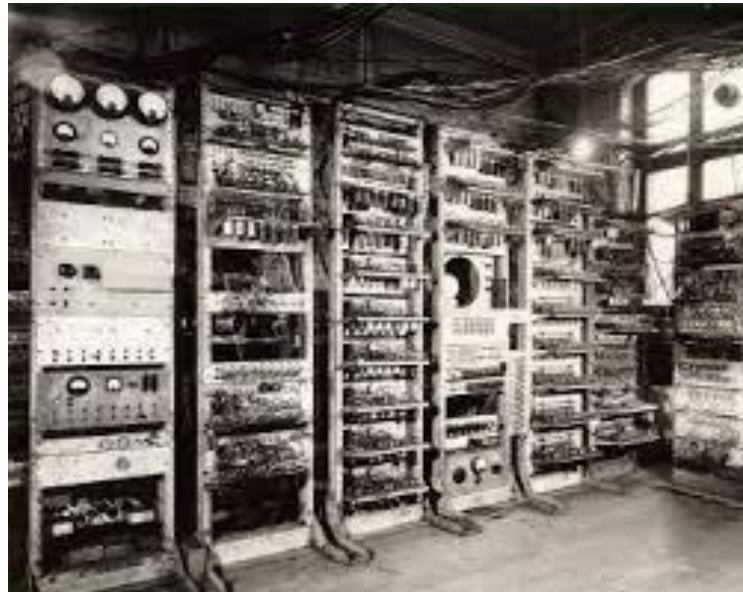
- In 1890 Herman Hollerith invented the recording of data on a medium that could then be read by a machine.
- After some initial trials with paper tapes, he settled on punch cards, developing punched card data processing technology for the 1890 US census.
- Influenced by Babbage's work.
- Began Tabulating Machine Company which is later changed its name as TBM.

- By using punch cards, Hollerith created a way to store and retrieve information and this was the first type of read and write technology.

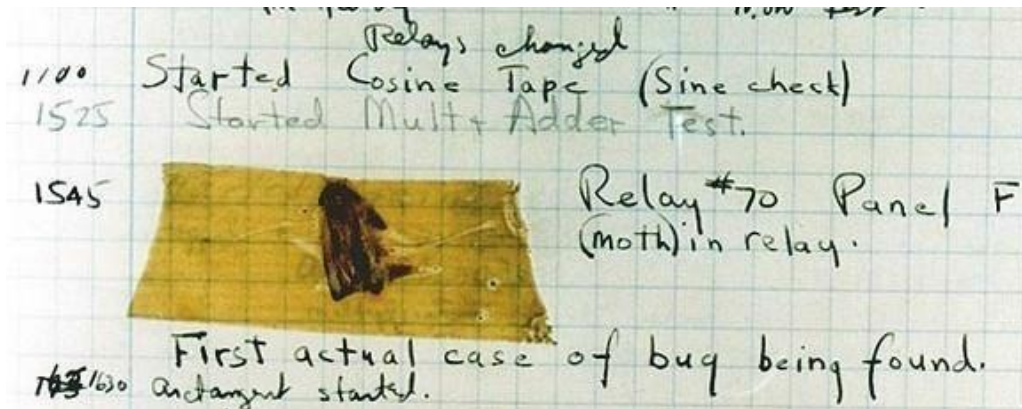


❑ Mark 1

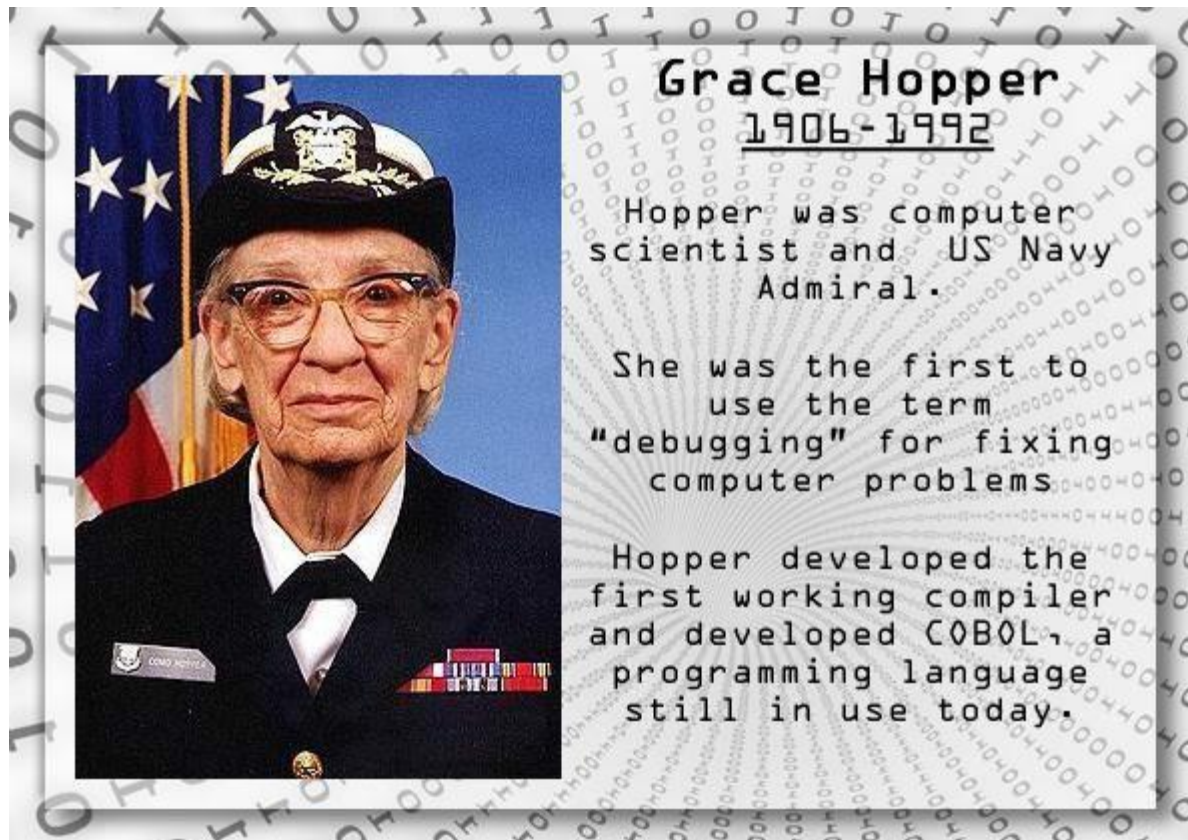
- A professor of Physics at Harvard University, **Howard Aiken** was supported by IBM to build **Mark 1** in 1944.
- Digital device.



- One of the primary programmers for the Mark 1 was a woman, **Grace Hopper**.
- One day, the program she was running gave incorrect results and, upon examination, a moth was found blocking one of the relays.
- The **bug was removed** and the program performed to perfection. Since then, a program error in computer has been called a bug.



- Debugging is a process of finding and correcting errors, in a computer program or a piece of electronic hardware.



- As the time passed, the device of more suitable and reliable machine was needed which could perform our work more quickly.
- In 1946, tiny electronic pathways called circuits began to perform the counting that had formerly been done by gears and other mechanical parts.
- The first successful electronic computer called ENIAC was developed in 1946 and it was the starting point of the current generation of computer.

The Computer Generations

- Generation means state of an improvement in the development of a product.
- With each new generation, the circuitry has gotten smaller and more advance than the previous generation before it.
- As a result of the miniaturization, speed, power and memory of computers has proportionally increased.
- New discoveries are constantly being developed.

First Generation of Computers (1946-1958)

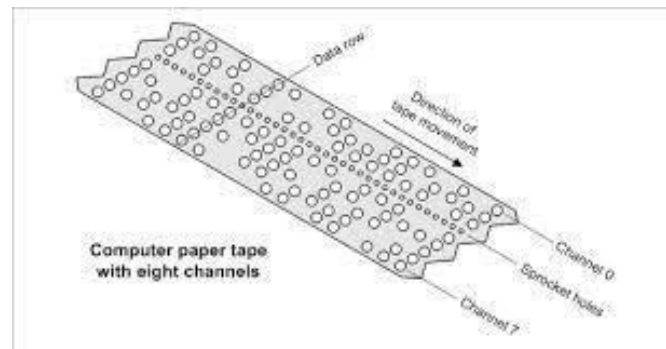
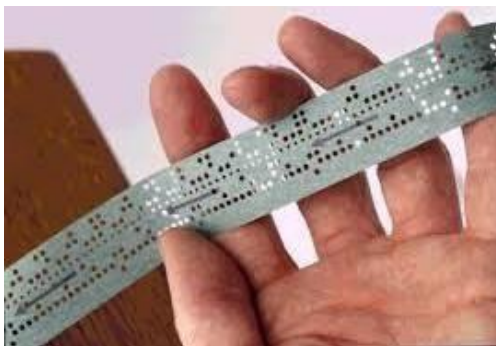
- Initially in the first generation computer the concept of vacuum tube was used.
- A vacuum tube was such an electronic component which had very less work efficiency and so it could not work properly and it required a large cooling system.



- Have used **magnetic drums** for memory.
 - Metal cylinder coated with magnetic iron-oxide material on which data and program can be stored.
 - As the drum rotates at a speed of up to 3,000 rpm, the device's read write heads deposit magnetized spots on the drum during the write operation and sense these spots during a read operation.



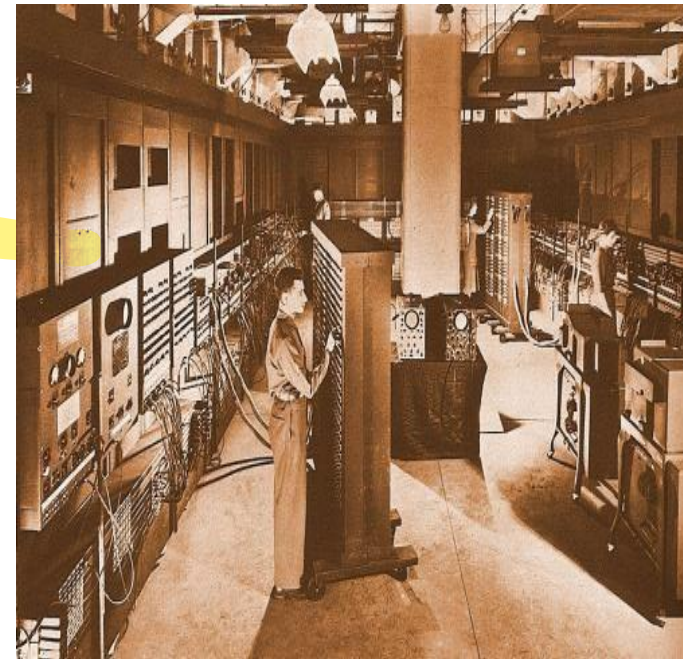
- Relied on **machine language** (0s and 1s) - the lowest level programming language understood by computers, to perform operations.
- Could **only solve one problem at a time.**
- Input was based on punched cards and paper tapes.
 - Record the data on a paper tape by punching rows of holes across the width of the tape.
 - The pattern of the holes in each data row represented a single data value or character.



The ENIAC and UNIVAC computers are examples.

ENIAC

- The ENIAC (Electro Numerical Integrator and Computer) was the first general purpose electronic computer built by Presper Eckert and Jhon Mauchly.
- It was the beginning of first generation computer.



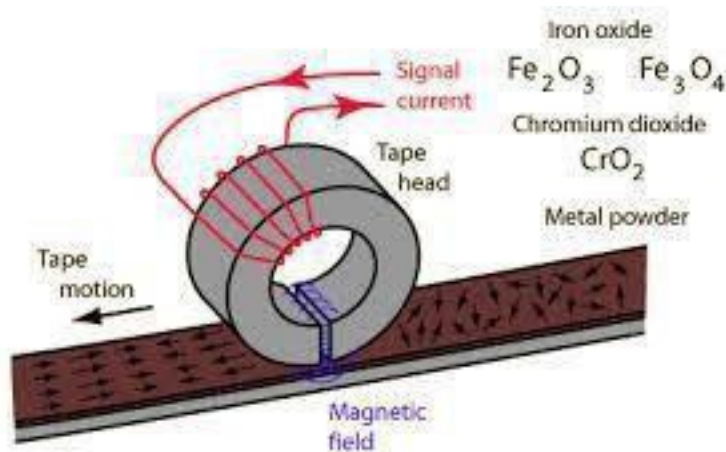
- ENIAC was huge and big computer and its weight was 30 tones and covered 150 square meters.
- It could store only limited or small amount of information.
- ENIAC in one second can complete 5000 additions, 257 multiplications and 38 divisions of numbers.
- The sponsor of the whole project was the American army which wanted to use ENIAC for making the first hydrogen bomb, information for weather forecast and some datum for the cosmic radiations.

UNIVAC

- UNIVAC (Universal Automatic Computer) was the first computer to be sold to businesses.
- Used to predict the presidential election of Dwight Eisenhower.



- Contained 5400 vacuum tubes and used magnetic tapes to give instructions to the computer.
 - Magnetic tape is a sequential storage medium used for data collection.
 - Tape is made of flexible plastic with one side coated with a ferromagnetic material.



Issues

- Huge
 - Thousands of vacuum tubes were there in a machine
- Slow
 - As much heat was generated
- Expensive
 - Heat that had to be cooled by gigantic air conditioners

Second Generation of Computers (1956 - 1963)

- The year 1948, Transistors was invented by William Shockley, John Bardeen, Walter Brattain.
- Transistor was a three legged, reliable, tiny, cheaper electronic component which shrunk the size of the first generation computers.
- Had greater computational speed, required no warm-up time and consumed far less electricity.
- Each replaced 40 vacuum tubes.



- Moved from binary machine language to symbolic or assembly languages which allowed programmers to specify instructions in words.
- High level programming languages, such as COBOL and FORTRAN were developed.
- First computers that stored their instructions in their memory, magnetic core technology.
- Punch cards for input.
- IBM's 7000 Series, TRADIC, TX-0, LARC and MARK III are the examples of second generation computers.

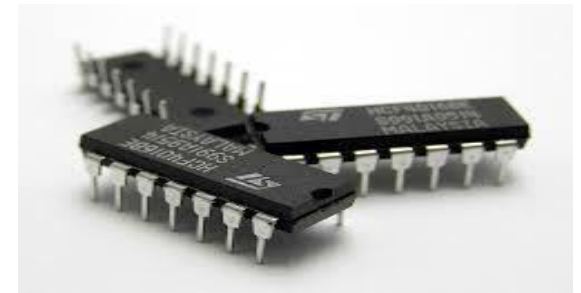
Advantages

- Fast
- More reliable
- Small
- More energy efficient
- Much cheaper to build than a vacuum tube



Third Generation of Computers (1964 - 1971)

- The third generation computers emerged with the development of **Integrated Circuits (IC)** also known as semiconductor chips.
- IC was invented by **Robert Noyce** and **Jack Kilby** in 1958-59.
- IC is a single component containing a number of transistors – Packs a huge number of transistors onto a single wafer of silicon or semiconductor material.



- Vastly increased the power of a single computer and lowered its cost.
- Number of transistors that can be placed on a single chip has doubled every two years.
- Most electronic devices today also use the integrated circuits built in to boards.
- Size of the third generation machines dropped to the size of small file cabinets and could carry out instructions in billionths of a second.
- Used keyboards and monitors as input and output devices and interfaced with an operating system.

- Operating system allowed the device to run many different applications at one time with a central program that monitored the memory.
- PDP-8, CDC 7600, INTEL 4004, IBM 360 and IBM 370 are the examples of third generation computers.



Fourth Generation of Computers (1971-Today)

- Marked the use of microprocessor.
 - A **microprocessor** is a computer processor that incorporates the functions of a computer's central processing unit (CPU) on a single integrated circuit or at most a few integrated circuit.
 - **4004 chip** was the **first microprocessor** introduced by Intel Corporation.



- Object oriented programming languages such as Visual Basic, and JAVA are characteristic of this computer generation.
- Many high-level languages were developed in this generation such as COBOL,FORTRAN,BASIC,PASCAL and C language.
- Networking between the systems was developed.
- IBM 4341, DEC 10, STAR 1000, PUP 11 APPLE II are the examples of fourth generation computers.

Today Computers

- Is classified as fourth generation computers.
- Faster, more powerful, tremendous data storage and processing capacity.
- New brands and models would come out the market almost every other month.
- Computers become more affordable.
- Computers can now be found in schools, offices even in homes.
- There has been a tremendous improvement in software technology.
 - Different software applications to choose from: word processing, spreadsheets, database management, games and entertainment.

Fifth Generation of Computers (Present & Beyond)

- Based on artificial intelligence and natural languages, are still in development, though there are some applications, such as voice recognition, that are being used today.
- The use of parallel processing and superconductors is helping to make artificial intelligence a reality.
- Quantum computation and molecular and nanotechnology will radically change the face of computers in years to come.

- The goal of fifth generation computing is to develop devices that respond to **natural language** input and are capable of learning and self-organization.



Summary

Generation	Time Period	Technology	Machine / Device
Pre Generation	1623-1900	Gears, Mechanical parts	Abacus, Napier's bones, Pascaline, Difference Engine and Analytical Engine
First Generation	1946-1957	Vacuum Tubes	ENIAC, UNIVAC
Second Generation	1956-1963	Transistors	TRADIC, TX-0, IBM 7000 Series, MARK III
Third Generation	1964-1971	IC	PDP-8, ICL 2900, IBM 360
Fourth Generation	1971-Present	Microprocessor	IBM 4341, DC 10, STAR 1000
Fifth Generation	Present- Beyond	AI, Natural Languages	

THANK YOU

